

Ex-3 Prompt Engineering Techniques – Implementation with Examples

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Aim

To design and implement prompt engineering techniques for building an AI-powered chatbot that assists users in product troubleshooting, order tracking, and general inquiries.

Introduction

Prompt Engineering is the process of designing effective prompts to obtain accurate and meaningful responses from AI systems. Different prompting techniques help improve chatbot communication, response quality, and user interaction.

In this experiment, various prompt engineering techniques are implemented using chatbot-based customer support examples such as order tracking, troubleshooting, and inquiry handling.

Algorithm

1. Identify the chatbot use case (customer support).
2. Select different prompt engineering techniques.
3. Design prompts for each technique.
4. Execute prompts using AI tools (ChatGPT / Gemini).

5. Analyze the outputs generated.
6. Compare responses for effectiveness.
7. Document examples and outputs.
8. Prepare report and convert to PDF.
9. Submit through GitHub.

AI Tools Used

- OpenAI – ChatGPT
- Google – Gemini

Prompt Engineering Techniques with Examples

1. Straightforward Prompt

Description

Straightforward prompting gives direct instructions to the AI system. The response generated is usually simple, clear, and concise.

Prompt

Explain how to track an online order.

ChatGPT Output

1. Visit the shopping website or application.
2. Login using your account credentials.
3. Open “My Orders”.
4. Select the order you want to track.
5. Click on “Track Order”.
6. View the current delivery status and estimated delivery date.

Gemini Output

To track an online order:

- Open the shopping platform.

- Go to your orders section.
- Enter or select the order ID.
- Click the tracking option to view shipment updates.

Analysis

Feature	ChatGPT	Gemini
Clarity	Very clear	Clear
Detail Level	More detailed	Short and concise
User Friendliness	High	Medium

2. Tabular Format Prompting

Description

Tabular prompting organizes information in rows and columns, making responses easier to read and compare.

Prompt

Provide customer issues and solutions in a table format.

ChatGPT Output

Customer Issue	Solution
Order delay	Check tracking ID
Product defect	Request replacement
Payment failure	Retry payment or contact support
Wrong product received	Apply for return

Gemini Output

Issue	Suggested Solution
Delayed delivery	Verify tracking details
Damaged product	Ask for replacement

Refund issue	Contact customer support
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Analysis

Feature	ChatGPT	Gemini
Table Organization	Excellent	Good
Number of Examples	More	Moderate
Readability	High	High

3. Missing Word Prompting

Description

Missing word prompting helps the AI predict or complete missing information in a sentence.

Prompt

Fill in the blank: The order status can be checked using _____.

ChatGPT Output

tracking ID / order ID

Gemini Output

tracking number

Analysis

Feature	ChatGPT	Gemini
Accuracy	High	High
Variety of Answers	Multiple answers	Single answer
Relevance	Excellent	Good

4. Preceding Question Prompting

Description

In preceding question prompting, the AI first asks a clarifying question before generating the final response.

Prompt

Before answering, ask what the issue is. Then provide a solution.

ChatGPT Output

AI: What issue are you facing with your order?

User: My order is delayed.

AI: Please check the tracking details in the app. If the delay continues, contact customer support for assistance.

Gemini Output

AI: Could you describe your issue?

User: I received a damaged product.

AI: You can request a replacement or refund through the return section.

Analysis

Feature	ChatGPT	Gemini
Interaction Quality	Excellent	Good
Conversational Flow	Natural	Simple
User Engagement	High	Moderate

Comparative Analysis of ChatGPT and Gemini

Parameter	ChatGPT	Gemini	Comparison (Visual)
 Response Detail	More detailed 	More concise 	
 Clarity	Excellent 	Good 	
 Conversational Ability	Strong 	Moderate 	
 Structured Output	Excellent 	Good 	
 User Interaction	Interactive 	Less interactive 	
 Overall Performance	Highly effective 	Effective 	

Output

A report demonstrating various prompt engineering techniques with chatbot-based customer support examples was successfully created and analyzed using AI tools.

Result

Thus, different prompt engineering techniques were implemented and analyzed successfully for AI-powered chatbot development.

Conclusion

The experiment demonstrated how different prompt engineering techniques improve chatbot communication and response quality. Straightforward prompting provided direct answers, tabular prompting improved readability, missing word prompting enhanced predictive

capability, and preceding question prompting improved interaction flow. Comparing ChatGPT and Gemini showed that both AI tools are effective, while ChatGPT produced more detailed and interactive responses.

References

1. [OpenAI ChatGPT](#)
2. [Google Gemini](#)
3. [Prompt Engineering Guide](#)
4. [OpenAI Documentation](#)